

pfSense Firewall Rules Guide for SOC Home Lab

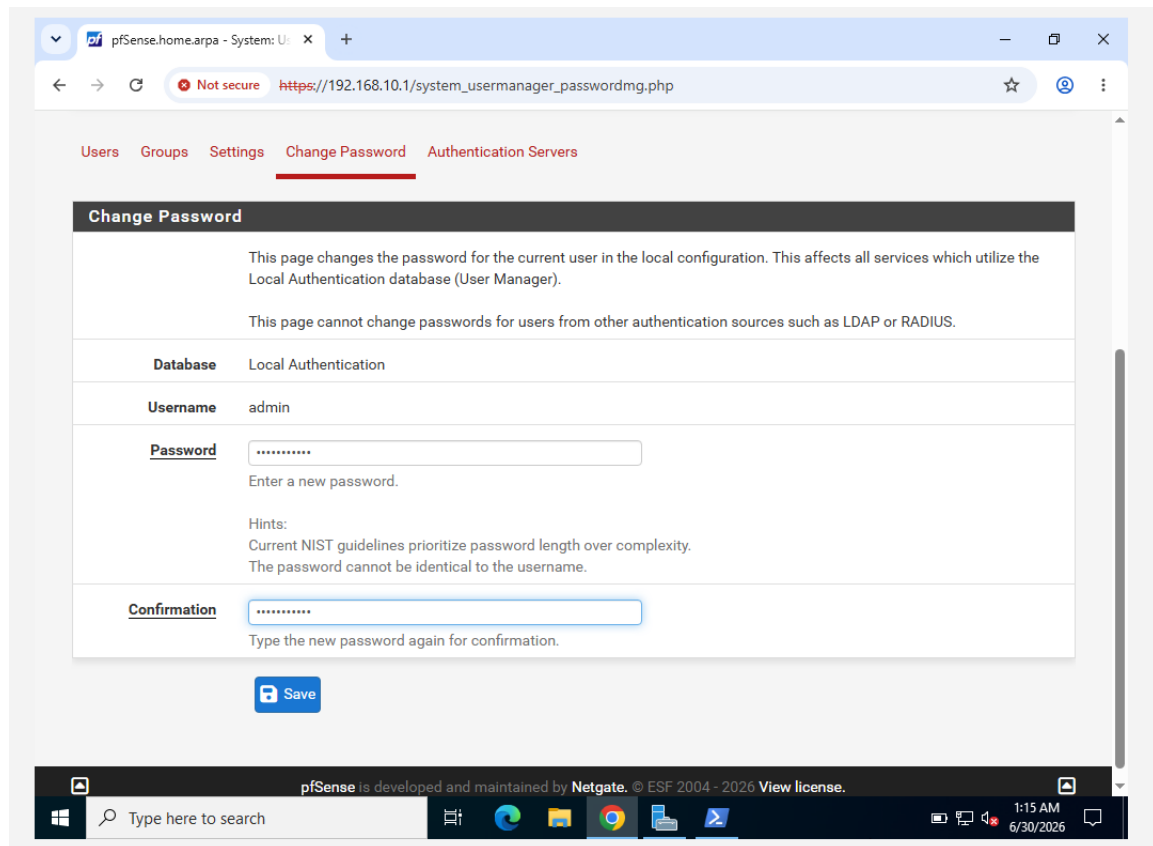
This documents guide user's to create rules of firewall for there SOC lab consisting of pfSense, Windows Server, Windows 10, and Wazuh.

LAN Rules

Rule 1 : Change the admin Password of Pfsense

Purpose: Change default Pfsense password for security.

Navigate: **System > User Manager > Users** tab. Click the **Edit**



The screenshot shows a web browser window with the URL `https://192.168.10.1/system_usermanager_passwordmg.php`. The page title is "Change Password". It contains a form with the following fields:

- Database:** Local Authentication
- Username:** admin
- Password:** A text input field with a placeholder "Enter a new password." and a hint: "Current NIST guidelines prioritize password length over complexity. The password cannot be identical to the username."
- Confirmation:** A text input field with a placeholder "Type the new password again for confirmation."

At the bottom of the form is a blue "Save" button. The footer of the page states: "pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license." The Windows taskbar at the bottom shows the date and time as 1:15 AM on 6/30/2026.

Rule 2: Allow Admin PC to pfSense Web UI

Purpose: Only the admin PC can manage pfSense.

Create: Firewall > Rules > LAN > Add.

Action: Pass

Protocol: TCP

Source: Address

Destination: Address

Destination Port: HTTPS (443)

Enable logging: Yes

Rule 3: Block pfSense Web UI from Other LAN Hosts

Purpose: Prevent other LAN devices from administering the firewall.

Action: Block

Protocol: TCP

Source: LAN subnet

Destination: LAN Address

Destination Port: HTTPS (443)

Enable logging: Yes

Rule 4: Allow DNS

Purpose: Clients query DNS.

Action: Pass

Protocol: TCP/UDP

Source: LAN subnet

Destination: This Firewall

Destination Port: 53 (DNS)

Rule 5: Allow HTTP/HTTPS

Purpose: Windows updates, package downloads, browsing.

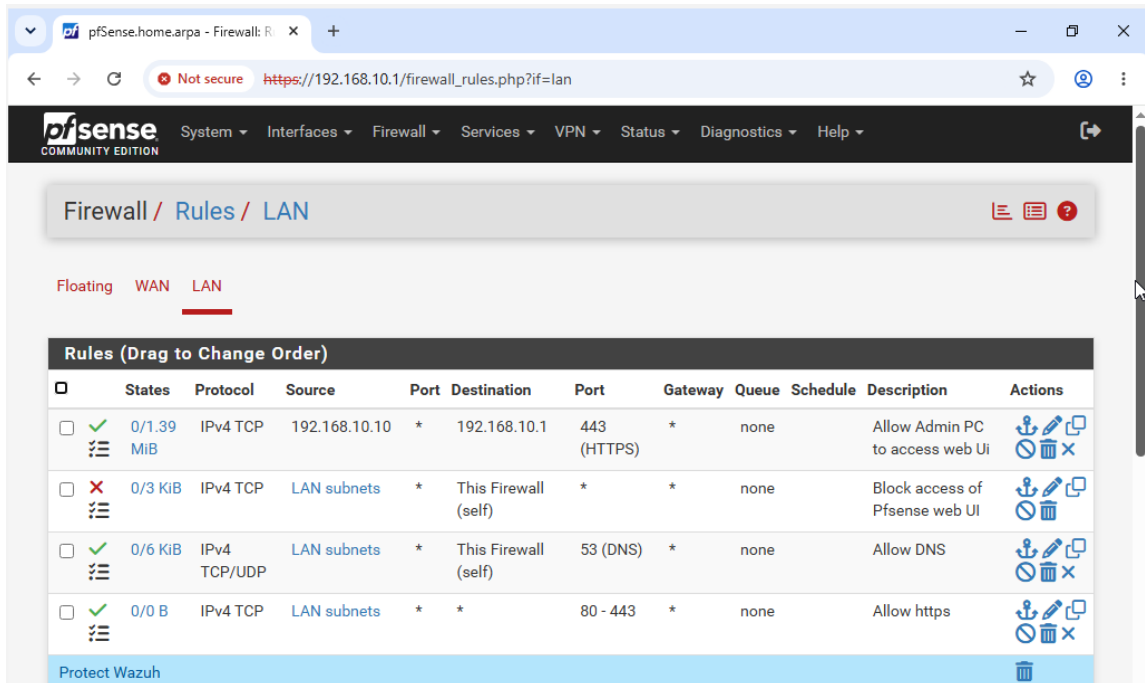
Action: Pass

Protocol: TCP

Source: LAN subnet

Destination: Any

Destination Ports: 80,443



Create a separator for Wazuh rules for better Understanding

Rule 6: Allow Wazuh Agents

Purpose: Agents communicate with Wazuh Manager.

Action: Pass

Protocol: TCP

Source: Windows Server, Windows 10

Destination: Wazuh Server (192.168.10.30)

Destination Ports: 1514-1515

Rule 7: Allow only admin PC to access wazuh web UI

Purpose: only admin PC have access to wazuh

Action: Pass

Protocol: TCP

Source: Windows Server

Destination: Wazuh Server (192.168.10.30)

Destination Ports: 443

Rule 8: Block wazuh Web UI from Other LAN Hosts

Purpose: Prevent other LAN devices from administering the wazuh.

Action: Block

Protocol: TCP

Source: LAN subnet

Destination: Wazuh Server (192.168.10.30)

Destination Ports: 443

Protect Wazuh											
☐		0/0 B	IPv4 TCP	192.168.10.10	*	192.168.10.30	1514 - 1515	*	none	Allow WinServer to send log to wazuh	     
☐		0/0 B	IPv4 TCP	192.168.10.104	*	192.168.10.30	1514 - 1515	*	none	Allow Win10 to send log to wazuh	     
☐		0/0 B	IPv4 TCP	192.168.10.10	*	192.168.10.30	443 (HTTPS)	*	none	Block access of wazuh webUI	     
☐		0/0 B	IPv4 TCP	LAN subnets	*	192.168.10.30	443 (HTTPS)	*	none	Block access of wazuh web UI	     

Create a separator for Window Server rules for better Understanding

Rule 9: Allow Active Directory

Purpose: Domain authentication.

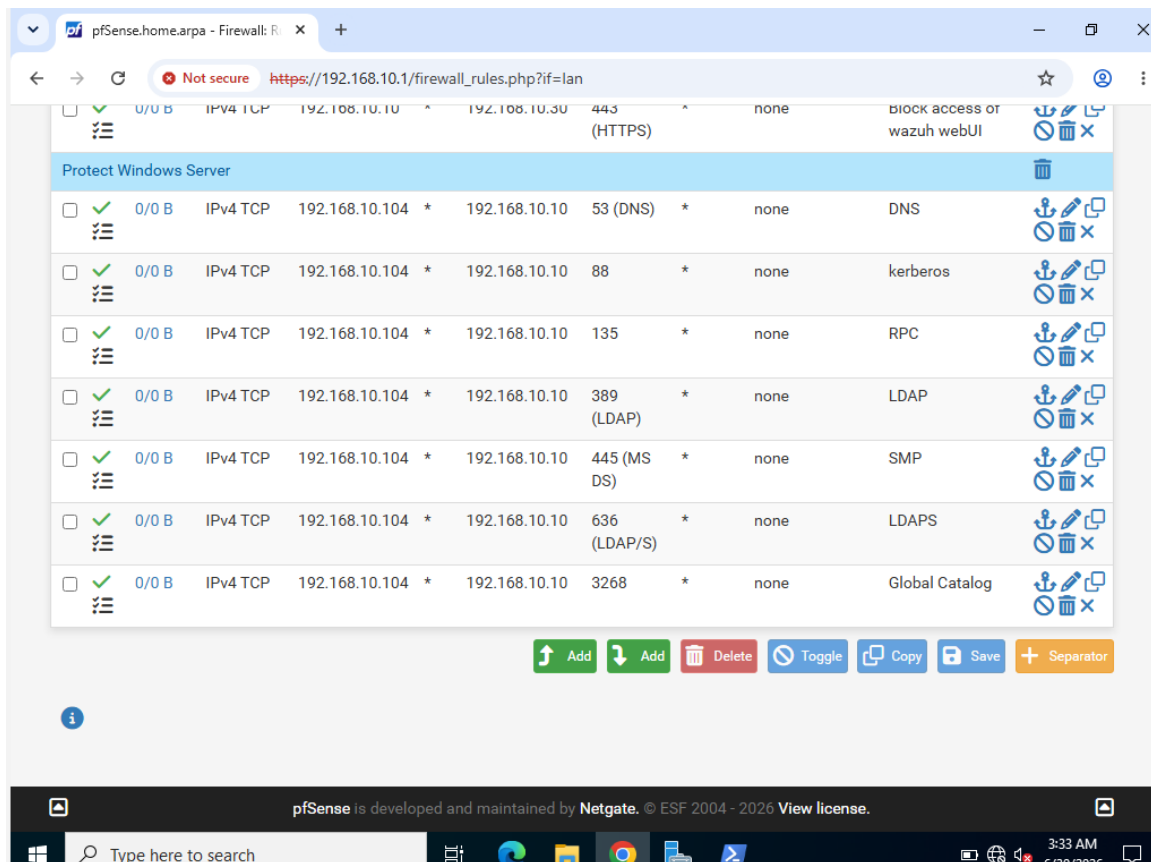
Action: Pass

Protocol: TCP/UDP

Source: Windows 10

Destination: Windows Server

Ports: 53,88,135,389,445,636,3268



Rule 10: Default Deny

Purpose: Any traffic not matching an allow rule is automatically blocked by pfSense. No manual rule required.

Rule 11: Log Important Rules

Purpose: Visibility for investigations and Wazuh integration.

Edit each important rule and enable 'Log packets that are handled by this rule'.

Recommended Rule Order

1. Allow Admin PC to pfSense
2. Block other pfSense Web UI access
3. Allow DNS
4. Allow HTTP/HTTPS
5. Allow Wazuh
6. Allow AD
7. pfSense built-in Default Deny

WAN Rules

Rule 1: Block Pfsense Web UI from Other WAN Hosts

Purpose: Prevent other WAN devices from administering the firewall.

Action: Block

Protocol: TCP

Source: WAN subnet

Destination: address (192.168.10.1)

Destination Port: HTTPS (443)

Enable logging: Yes

Rule 2: Block external device to internal device

Purpose: Prevent other WAN devices to LAN device.

Action: Block

Protocol: TCP

Source: WAN subnet

Destination: LAN subnet

Destination Port: any

Enable logging: Yes

pfSense.home.arpa - Firewall: Rules

Not securehttps://192.168.10.1/firewall_rules.php?if=wan

pfSense
COMMUNITY EDITION

System ▾ Interfaces ▾ Firewall ▾ Services ▾ VPN ▾ Status ▾ Diagnostics ▾ Help ▾

Firewall / Rules / WAN

Floating **WAN** LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0/0 B	*	RFC 1918 networks	*	*	*	*	*		Block private networks	
☒	0/0 B	*	Reserved Not assigned by IANA	*	*	*	*	*		Block bogon networks	
☐	☒ 0/0 B	IPV4 TCP	WAN subnets	*	195.168.10.1	443 (HTTPS)	*	none		Block access of pfsense web UI	
☐	☒ 0/0 B	IPV4 TCP	WAN subnets	*	LAN subnets	*	*	none		Block external device to internal device	

↑ Add

↓ Add

Delete

Toggle

Copy

Save

Separator

Type here to search

3:39 AM
6/30/2026